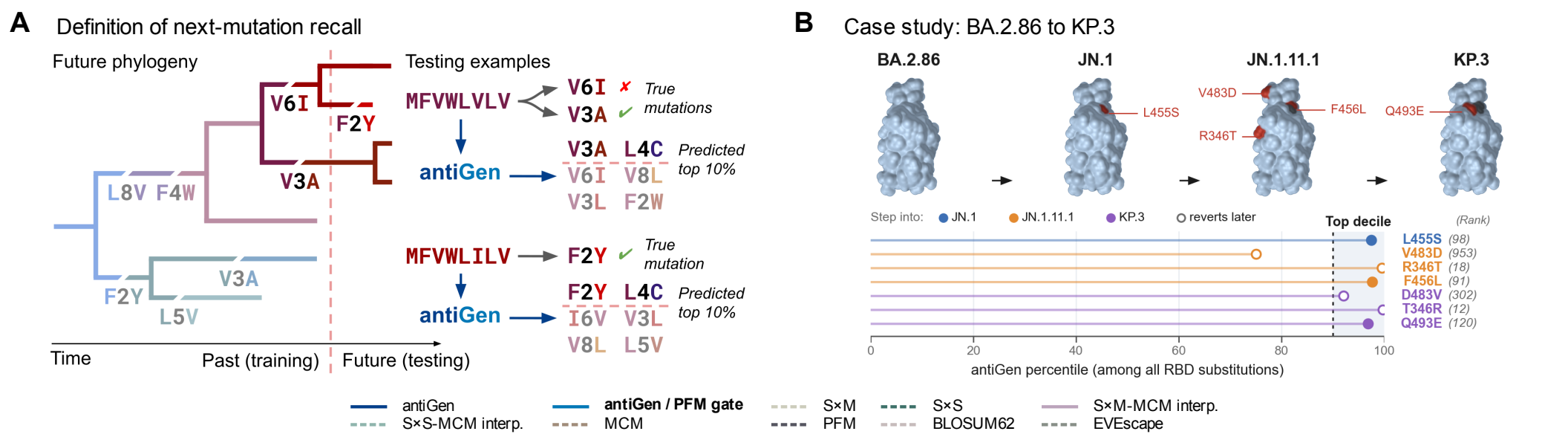

A
Definition of next-mutation recall
Future phylogeny
Testing examples
V6I
F2Y
V3A
L8V F4W
F2Y
V3A
L5V
MFVWLVLV
V6I ✗ True mutations
V3A ✓
antiGen
V3A L4C Predicted top 10%
V6I V8L
V3L F2W
MFVWLILV
F2Y ✓ True mutation
antiGen
F2Y L4C Predicted top 10%
I6V V3L
V8L L5V
Time
Past (training)
Future (testing)
B
Case study: BA.2.86 to KP.3
BA.2.86
JN.1
JN.1.11.1
KP.3
L455S
V483D
F456L
R346T
Q493E
Step into: JN.1 JN.1.11.1 KP.3 reverts later
Top decile
(Rank)
L455S (98)
V483D (953)
R346T (18)
F456L (91)
D483V (302)
T346R (12)
Q493E (120)
0
20
40
60
80
100
antiGen percentile (among all RBD substitutions)
antiGen
S×S-MCM interp.
antiGen / PFM gate
MCM
S×M
PFM
S×S
BLOSUM62
S×M-MCM interp.
EVEscape
C
Top decile recall, all RBD cells (per cell)
P(cell in top decile)
0.0
0.2
0.4
0.6
2021
2022
2023
2024
Train / test cutoff
D
Top decile recall, all RBD cells (per mutation)
P(obs. mut. in top decile)
0.00
0.25
0.50
0.75
2021
2022
2023
2024
Train / test cutoff
Model type
antiGen
antiGen gated hybrid
Statistical models
Deep Mutational Scans
Pandemic Data in Training Window
Pre-Pandemic Data Only
Designed in this work
External
E
All RBD, per cell
F
All RBD, per mutation
G
De novo RBD, per cell
H
De novo RBD, per mutation
All RBD, 2023 cutoff
antiGen (CV x10)
antiGen / PFM gate
S×M
S×S
S×M-MCM interp.
S×S-MCM interp.
Codon-B62 mixture
PFM
PFM Conservation
MSA Transformer
MSA Trans. (BA.2 eval)
MSA Trans. (XFG eval)
EVE
MCM
BA.1 DMS (bind)
BA.1 DMS (expression)
BA.2 DMS (bind)
BA.2 DMS (expression)
BA.2 DMS (LY-CoV1404 escape)
EVEscape
ESM 2 (35M), untrained
ESM 2 (650M), untrained
BLOSUM62
Codon table
1st
2nd
3rd
0.1
0.2
0.3
0.4
0.5
0.6
0.25
0.50
0.75
Recall at q = 0.1
I
All escape, per cell
J
All escape, per mutation
K
De novo escape, per cell
L
De novo escape, per mutation
Escape, 2021 cutoff
antiGen (CV x10)
antiGen / PFM gate
S×M
S×S
S×M-MCM interp.
S×S-MCM interp.
Codon-B62 mixture
PFM
PFM Conservation
MSA Transformer
MSA Trans. (BA.2 eval)
MSA Trans. (XFG eval)
EVE
Tranception
MCM
BA.1 DMS (bind)
BA.1 DMS (expression)
BA.2 DMS (bind)
BA.2 DMS (expression)
BA.2 DMS (LY-CoV1404 escape)
EVEscape
ESM 2 (35M), untrained
ESM 2 (650M), untrained
BLOSUM62
Codon table
1st
2nd
3rd
0.1
0.2
0.3
0.4
0.5
0.2
0.4
0.6
0.8
Recall at q = 0.1